

Patient Shortage in Histopathology: Do Coverage and Patient Similarity Explain the Breadth Benefit?

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Abstract

TCGA-UT patch classifiers trained on more patients are more accurate than redundancy-adjusted effective support predicts, and at a fixed patient count a cohort of mutually similar patients is 8.30 percentage points less accurate than a random cohort. Without training new classifiers, this analysis asks whether two cohort properties account for both findings: similarity between patients, added to effective support, and coverage, the distance from validation patients to their nearest training patient. On 135 random cohorts, adding coverage lowers the accuracy gained per doubling of patients beyond effective support from 2.62 to 0.79 points, but the 95% interval of -1.25 to 2.82 points is wide because coverage and patient count change together. A relation fitted to these random cohorts predicts a concentration damage of 6.80 points against the observed 8.30, and a gain from coverage-maximizing selection of 2.10 points against the observed 1.04; both prediction errors exclude zero but cross the one-point threshold. The outcome is inconclusive. Coverage and similarity are compatible with the whole breadth benefit and with most of the concentration damage, but the random cohorts do not identify their contributions precisely.

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1 Motivation

At 160 patches per TCGA-UT class, twenty patients contributing eight patches each outperform five patients contributing 32 patches each by 9.52 percentage points of patient-macro balanced accuracy (Queisler, 2026b). Effective support, which discounts patches by their correlation within a patient, explains less than half of this gap: each doubling of patients still adds 2.62 points at equal effective support, about 5.2 points over the two doublings (Queisler, 2026c). Doubling tissue source sites at a fixed patient count adds 1.06 points (Queisler, 2026d).

Cohort composition matters more. Among cohorts of twenty patients with eight patches each, a cohort built from one patient and its nineteen nearest neighbours in the Virchow2 embedding is 8.30 points less accurate than a random cohort (95% interval: 7.66 to 8.95), and a cohort chosen by coverage-maximizing selection is 1.04 points more accurate (0.44 to 1.66) (Queisler, 2026a). The damage falls on test patients far from the clustered cohort. Two properties change together in that experiment, however. The clustered cohort covers the class’s between-patient variation poorly, and its patients resemble one another, so they carry fewer independent observations than effective support counts. This report asks: *do coverage and similarity between patients together account for the benefit of additional patients in random cohorts, and does the relation that describes random cohorts also predict the effect of cohort composition?*

2 Study design

No classifier is trained. The analysis uses the stored predictions of two experiments that share the 30 eligible TCGA-UT cancer types, the frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), the three fixed patient-disjoint splits with their natural validation and test partitions, and multinomial logistic regression with validation-selected regularization. The *random cohorts* are the nine cells of the breadth–depth grid, $G \in \{5, 10, 20\}$ patients per class with $m \in \{8, 16, 32\}$ patches each, in three splits and five draws: 135 cohorts (Queisler, 2026b). The *composition cohorts* are the clustered, random, and dispersed allocations of Queisler (2026a) at $G = 20$ and $m = 8$, again in three splits and five draws: 45 cohorts. A cohort is one split, allocation, and draw, and its accuracy A is patient-macro balanced accuracy in percent over all 30 classes.

Two quantities describe each cohort (Table 1). Both use training and validation patients only, and both are computed per class and split and then averaged over classes.

Quantity	What it measures	Random against clustered cohort
Similarity-adjusted effective support N_{eff}^{ω}	Independent observations, counting correlation within and between patients	Equal to effective support on average; lower for similar patients
Coverage distance r	Distance from validation patients to their nearest training patient	Small; large when patients form one neighbourhood

Table 1: The two cohort quantities separate redundancy from coverage. In random cohorts both change mainly with the patient count; at a fixed patient count, cohort composition moves both.

2.1 Similarity-adjusted effective support

Effective support of class c uses the full-feature intraclass correlation $\bar{\rho}_c$ of Queisler (2026c), the share of feature variance that lies between patients. For split r , let $\bar{\mathbf{x}}_i$ be the mean training feature vector of patient i for class c , $\boldsymbol{\mu}_{rc}$ the average of these means over the class’s eligible training patients, and τ_{rc}^2 the between-patient variance of the class, estimated as $(B - W)/\tilde{m}$ from the mean squares of that report. The *between-patient similarity* of a cohort S with G patients is

$$\omega_{rc}(S) = \frac{1}{G(G-1)\tau_{rc}^2} \sum_{i \in S} \sum_{j \in S \setminus \{i\}} (\bar{\mathbf{x}}_i - \boldsymbol{\mu}_{rc})^\top (\bar{\mathbf{x}}_j - \boldsymbol{\mu}_{rc}), \quad (1)$$

the mean correlation between the deviations of two cohort patients from their class mean. Because the two patients in each term differ, sampling noise of the patient means does not enter the numerator in expectation. If patches of the same patient correlate at $\bar{\rho}_c$ and patches of two cohort patients at $\bar{\rho}_c \omega$, the variance of a class mean over Gm patches yields

$$N_{\text{eff}}^\omega = \frac{1}{30} \sum_c \frac{Gm}{1 + (m-1)\bar{\rho}_c + m(G-1)\bar{\rho}_c \omega_{rc}(S)}. \quad (2)$$

For randomly drawn patients, ω is close to zero and Equation (2) reduces to the effective support of Queisler (2026c). For patients drawn from one neighbourhood, ω is positive and the added term grows with both G and m .

2.2 Coverage distance

Coverage follows Queisler (2026a). The distance $d(i, j)$ between two patients is the cosine distance between their patient-mean embeddings after subtracting the average over classes. The coverage distance of cohort S is the mean distance from the class’s validation patients $\mathcal{V}_{rc}$ to their nearest cohort patient, averaged over classes,

$$r(S) = \frac{1}{30} \sum_c \frac{1}{|\mathcal{V}_{rc}|} \sum_{i \in \mathcal{V}_{rc}} \min_{j \in S} d(i, j). \quad (3)$$

3 Evaluation

The evaluation poses the two questions of Section 1 in turn: first on the random cohorts alone, then as a prediction for the composition cohorts.

3.1 Residual breadth benefit

Three linear regression models are fitted by least squares to the 135 random cohorts, each with split-specific intercepts α_r :

$$\begin{aligned} \text{(a)} \quad & A = \alpha_r + \beta \log N_{\text{eff}} + \gamma \log G, \\ \text{(b)} \quad & A = \alpha_r + \beta \log N_{\text{eff}}^\omega + \gamma \log G, \\ \text{(c)} \quad & A = \alpha_r + \beta \log N_{\text{eff}}^\omega + \delta r + \gamma \log G. \end{aligned} \quad (4)$$

Model (a) repeats the analysis of Queisler (2026c) at the cohort level, model (b) adds similarity between patients, and model (c) adds coverage. In each, the *residual breadth benefit* $b = \gamma \log 2$ is the accuracy gained per doubling of patients beyond the other terms. The primary estimate is b of model (c). The residual standard deviation of each model, and of model (c) without $\log G$, is reported alongside.

3.2 Prediction of cohort composition

Model (c) without $\log G$ is fitted to the 135 random cohorts only and applied to the 45 composition cohorts, whose N_{eff}^ω and r are computed in the same way. Averaging predicted accuracies over splits and draws gives the *predicted concentration damage* $\hat{\Delta}_{\text{con}} = \hat{A}_{\text{ran}} - \hat{A}_{\text{clu}}$ and the *predicted selection gain* $\hat{\Delta}_{\text{sel}} = \hat{A}_{\text{dis}} - \hat{A}_{\text{ran}}$. Their *prediction errors* are the differences from the observed contrasts, 8.30 and 1.04 points. The composition cohorts break the link between patient count and cohort quantities that holds in random cohorts, so this step tests whether the relation describes the quantities or only the patient count. The predicted accuracy of the random composition allocation, set against its observed 71.29 percent, checks the calibration of the relation at the same cohort size.

3.3 Uncertainty

Each of 2,000 paired Bayesian bootstrap replicates assigns $\text{Dirichlet}(1, \dots, 1)$ weights to the distinct test patients (Rubin, 1981), shared across both experiments, all cohorts, and all classes. Accuracies, models, predictions, and prediction errors are recomputed per replicate, and the 2.5th and 97.5th percentiles give exploratory 95% intervals. Correlations, similarities, and coverage distances are held fixed. The three splits overlap and are not independent replications.

4 Interpretation criteria

Conclusions rest on the test-patient intervals for b of model (c) and for both prediction errors, with the practical threshold of one percentage point used in the preceding experiments (Table 2). The two prediction errors carry unequal weight. The observed selection gain is itself close to one point, so a prediction within one point of it is weak evidence; the concentration damage carries the test.

Reading	Claim	Residual b , model (c)	Prediction errors
Coverage and similarity	Both quantities account for the breadth benefit and for cohort composition	Interval within $[-1, 1]$	Both intervals within $[-1, 1]$
Random cohorts only	Both quantities describe random cohorts but not the effect of composition	Interval within $[-1, 1]$	Concentration interval entirely outside $[-1, 1]$
Breadth beyond both	Additional patients help beyond coverage and similarity	Interval above 1	Not required

Table 2: Each reading predicts a distinct combination of the residual breadth benefit and the prediction errors. Intervals are test-patient 95% intervals in percentage points. All other combinations, including intervals that cross a threshold, are inconclusive.

Coverage and similarity. If patient count adds nothing beyond both quantities and the relation reproduces the concentration damage, coverage and similarity between patients are

candidate label-free signals of patient shortage. The relation then yields a forecast for coverage-maximizing selection against random sampling at five patients with 32 patches each, the allocation in which patient shortage was first observed. The forecast and its interval are recorded before any classifier of that experiment is trained.

Random cohorts only. If both quantities absorb the breadth benefit but miss the concentration damage by more than one point, they covary with patient count in random cohorts without capturing what makes a cohort informative. A forecast for selected cohorts would not be justified.

Breadth beyond both. If the residual stays above one point, additional patients contribute information that neither patient-mean coverage nor patient-mean similarity expresses. The next candidate would be a distance between the patch distributions of patients.

5 Results

Adding coverage distance to effective support lowers the residual breadth benefit from 2.62 to 0.79 percentage points per doubling of patients, but its 95% interval, -1.25 to 2.82 points, crosses both thresholds. The relation fitted to the random cohorts predicts a concentration damage of 6.80 points against the observed 8.30, a prediction error of -1.50 points (-2.61 to -0.40), and a selection gain of 2.10 points against the observed 1.04, an error of 1.06 points (0.23 to 1.88). Neither the residual nor the concentration error meets the conditions of any reading in Table 2, so the outcome is inconclusive.

5.1 Cohort quantities

The two quantities behave as intended in the composition cohorts (Table 3). Clustered cohorts have a mean between-patient similarity of $\omega = 0.241$, between 0.20 and 0.28 in every split and draw, which lowers similarity-adjusted effective support from 36.58 to 10.33, about the value of five random patients with eight patches each. Random and dispersed cohorts keep ω near zero and N_{eff}^{ω} near 53. Similarity varies strongly between cancer types: the clustered cohorts of uterine carcinosarcoma, adrenocortical carcinoma, pheochromocytoma and paraganglioma, mesothelioma, and uveal melanoma have ω below 0.07, and those of testicular germ cell tumours reach 0.63. The five types with the smallest ω include four of the five types in which clustering changed coverage least (Queisler, 2026a).

In random cohorts, ω is not zero but slightly negative, between -0.034 and 0.004 across cancer types at twenty patients. This range is consistent with the expectation $-1/(|\mathcal{E}_{rc}| - 1)$ for patients drawn without replacement from a pool of 30 to 514 patients whose mean defines the deviations in Equation (1). The term $m(G - 1)\bar{\rho}_c\omega$ multiplies this small value, so N_{eff}^{ω} exceeds N_{eff} by 16 percent for five patients with eight patches and by 56 percent for twenty patients with 32 patches.

Coverage distance depends almost entirely on the patient count. Across the fifteen random cohorts of one patient count, r ranges from 0.371 to 0.398 at five patients, from 0.329 to 0.338 at ten, and from 0.288 to 0.301 at twenty, whereas the means at five and twenty patients differ by 0.093. The composition cohorts partly leave this range. Clustered cohorts lie between the random cohorts of five and ten patients in coverage distance, but their similarity, at least 0.20, far exceeds the largest value in any random cohort, 0.016. Dispersed cohorts cover validation patients better than any random cohort does.

Cohorts	Cell or allocation	A	N_{eff}	N_{eff}^{ω}	ω	r
Random	$G = 5, m = 8$	61.97	9.15	10.59	-0.010	0.386
	$G = 5, m = 16$	62.13	9.74	11.53	-0.010	0.386
	$G = 5, m = 32$	62.25	10.06	12.10	-0.010	0.386
	$G = 10, m = 8$	67.04	18.29	22.88	-0.013	0.333
	$G = 10, m = 16$	67.28	19.47	25.09	-0.013	0.333
	$G = 10, m = 32$	67.39	20.12	26.50	-0.013	0.333
	$G = 20, m = 8$	71.76	36.58	52.37	-0.011	0.293
	$G = 20, m = 16$	72.02	38.94	58.61	-0.011	0.293
	$G = 20, m = 32$	72.08	40.25	62.61	-0.011	0.293
Composition	Clustered	62.99	36.58	10.33	0.241	0.352
	Random	71.29	36.58	53.36	-0.012	0.294
	Dispersed	72.33	36.58	52.65	-0.001	0.269

Table 3: Clustering lowers similarity-adjusted effective support to the level of five random patients, whereas in random cohorts it exceeds effective support. A is patient-macro balanced accuracy in percent (Queisler, 2026b,a). All other columns are means over three splits and five draws of the class-averaged quantities of Section 2. Within a patient count, the random cohorts share their patients across depths, so ω and r do not change with m . Effective support N_{eff} depends only on G , m , and the intraclass correlations, so it is equal for the three composition allocations.

5.2 Residual breadth benefit

Model (a) reproduces the residual breadth benefit of Queisler (2026c) at the cohort level, $b = 2.62$ points per doubling (2.04 to 3.17; Table 4). Replacing N_{eff} by N_{eff}^{ω} in model (b) raises the residual to 3.71 points. Because N_{eff}^{ω} grows more steeply with patient count than N_{eff} , the model attributes a smaller share of the accuracy gain to effective support, with $\beta = 1.49$ against 3.32, and a larger share to patient count. Similarity between patients therefore absorbs nothing of the breadth benefit in random cohorts, as expected when ω is near zero.

Model	β	δ	$b = \gamma \log 2$	Residual SD
(a)	3.32 [2.56, 4.09]		2.62 [2.04, 3.17]	1.11
(b)	1.49 [0.14, 2.79]		3.71 [2.65, 4.83]	1.11
(c)	0.62 [-0.66, 1.88]	-78.0 [-118.5, -39.3]	0.79 [-1.25, 2.82]	1.02
(c) without $\log G$	1.10	-86.3		1.02

Table 4: Coverage distance lowers the residual breadth benefit to below one point, but with an interval that crosses both thresholds. Models are those of Equation (4), fitted to the 135 random cohorts with split-specific intercepts. β is the accuracy change per unit of $\log N_{\text{eff}}$ or $\log N_{\text{eff}}^{\omega}$, and δ the change per unit of coverage distance, both in percentage points. Brackets give test-patient 95% intervals. The coefficients of model (c) without $\log G$ are recovered from its two predicted contrasts and carry no intervals. The residual standard deviation is in percentage points.

Adding coverage distance in model (c) improves the fit, with the residual standard deviation falling from 1.11 to 1.02 points, and lowers the residual to $b = 0.79$ points. Each 0.01 decrease in coverage distance adds 0.78 points. Over the 0.093 by which twenty random patients cover validation patients better than five, coverage accounts for 7.26 points; the residual adds 1.58 points over two doublings and similarity-adjusted effective support 0.98 points at eight patches.

These parts sum to 9.82 points, close to the observed 9.80-point advantage of twenty over five patients at eight patches. Dropping $\log G$ leaves the residual standard deviation unchanged at 1.02 points.

The interval of b , -1.25 to 2.82 points, is more than three times as wide as in model (a). It excludes neither a residual above the one-point threshold nor one below -1 . The coefficient of coverage distance is also uncertain, with an interval from -118.5 to -39.3 . Both follow from the near-collinearity of r and $\log G$ described above: within one patient count, coverage distance varies over less than a third of the difference between five and twenty patients.

5.3 Prediction of cohort composition

Model (c) without $\log G$ is well calibrated for random cohorts of the composition experiment. It predicts 71.70 percent for the random allocation (70.36 to 72.92) against the observed 71.29 percent (69.93 to 72.51), a difference of 0.41 points.

Contrast	Observed	Predicted	Prediction error	Predicted part	
				Similarity	Coverage
Concentration damage	8.30	6.80 [5.90, 7.71]	-1.50 [-2.61 , -0.40]	1.81	4.99
Selection gain	1.04	2.10 [1.51, 2.72]	1.06 [0.23, 1.88]	-0.01	2.12

Table 5: The relation fitted to random cohorts reproduces most of the concentration damage but overpredicts the selection gain. Predictions apply model (c) without $\log G$ to the 45 composition cohorts. Observed contrasts are those of Queisler (2026a). The prediction error is the predicted minus the observed contrast. Brackets give test-patient 95% intervals. The predicted parts split each point prediction into the contributions of $\log N_{\text{eff}}^{\omega}$ and r ; they are descriptive and carry no intervals. All values are in percentage points.

The predicted concentration damage is 6.80 points (5.90 to 7.71), 82 percent of the observed damage (Table 5). The prediction error of -1.50 points has an interval of -2.61 to -0.40 , which excludes zero and crosses the threshold of -1 . Coverage contributes 4.99 points of the prediction and similarity 1.81 points. The predicted selection gain is 2.10 points (1.51 to 2.72), about twice the observed 1.04 points. It rests entirely on coverage, because random and dispersed cohorts differ little in similarity, and its error interval of 0.23 to 1.88 points also excludes zero and crosses the threshold.

The two errors point in opposite directions. The relation predicts too little accuracy loss where coverage is poor and similarity high, and too much gain where coverage is better than in any random cohort. A single linear slope in coverage distance, estimated from random cohorts, therefore misses the shape of both composition effects, although it captures their order and most of the larger one.

6 Discussion and conclusion

Coverage and similarity between patients are compatible with the whole breadth benefit of TCGA-UT random cohorts and with most of the effect of cohort composition, but this analysis does not establish that they account for either. In random cohorts, coverage distance takes over most of the accuracy gain that effective support leaves to patient count. The point estimate of the residual falls from 2.62 to 0.79 points per doubling, the fit improves, and dropping patient count from the model does not worsen the fit. The random cohorts, however, cannot tell coverage and patient count apart precisely: coverage distance is nearly fixed by the number of patients, so an interval of four points around the residual remains.

The prediction is the more demanding test, because the composition cohorts break the link between patient count and coverage. A relation that sees only random cohorts and no patient count predicts that twenty clustered patients lose 6.80 points against twenty random patients, and that coverage-maximizing selection gains 2.10. The observed contrasts are 8.30 and 1.04. Neither error is within the one-point threshold, and both deviate from linearity in a consistent way: accuracy falls faster than coverage predicts for the poorly covering and highly similar clustered cohort, and gains less than predicted for the dispersed cohort, which covers validation patients better than any random cohort. The second deviation fits diminishing returns from coverage once random sampling at twenty patients has covered most validation patients. The first may reflect a similarity effect larger than the relation assigns, since its coefficient for effective support is estimated from variation in patient and patch counts, in which similarity barely varies. It may also reflect the over-representation of one appearance of the cancer type, which neither quantity measures (Queisler, 2026a).

The contributions of coverage and patient count separate only when coverage varies at a fixed patient count, and so far that holds only at twenty patients. Comparing coverage-maximizing selection with random sampling at five patients with 32 patches each would add a second patient count at which coverage varies independently of the number of patients, where random cohorts cover validation patients at 0.386 and have more coverage left to gain. Such a comparison would test the breadth benefit directly and the shape of the coverage relation beyond the range of random cohorts.

7 Limitations

The logarithmic and linear terms of the models are assumed rather than tested, and a relation fitted to random cohorts carries no information about similarity values that random cohorts do not reach. Both quantities summarize a patient by its mean embedding and a class by scalar correlations. The composition cohorts share test patients with the random cohorts, so the prediction step is out of sample for training cohorts, not for evaluation patients. All results are conditional on Virchow2 features, multinomial logistic regression, and the three overlapping splits.

References

- Yannik Queisler. Patient shortage in histopathology: Does cohort composition matter at a fixed patient count? Technical report, TU Berlin, 2026a. Companion cohort-composition report.
- Yannik Queisler. Patient shortage in histopathology: Breadth, depth, and effective support. Technical report, TU Berlin, 2026b. Companion breadth–depth report.
- Yannik Queisler. Patient shortage in histopathology: Does full-feature redundancy explain the residual breadth benefit? Technical report, TU Berlin, 2026c. Companion full-feature redundancy report.
- Yannik Queisler. Patient shortage in histopathology: Is the residual breadth benefit a site-coverage benefit? Technical report, TU Berlin, 2026d. Companion site-coverage report.
- Donald B. Rubin. The bayesian bootstrap. *The Annals of Statistics*, 9(1):130–134, 1981. doi: 10.1214/aos/1176345338.
- Eric Zimmermann, Eugene Vorontsov, Julian Viret, Adam Casson, Michal Zelechowski, George Shaikovski, Neil Tenenholtz, James Hall, Thomas Fuchs, Nicolo Fusi, Siqi Liu, and Kristen Severson. Virchow2: Scaling self-supervised mixed magnification models in pathology. *arXiv preprint arXiv:2408.00738*, 2024. doi: 10.48550/arXiv.2408.00738.